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Ministry of Education

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College of Computer Sciences & Information Technology

Fitness Tracker

A Project of The Course:

Fundamentals of Web Programming

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Introduction:

The **Fitness Tracker** web application is designed to help users **manage** and **track their fitness** based on specific information entered by the user, such as **birth-date, weight, height,** and **activity level**.

The system supports **user sign-in** and **secure data storage**, allowing users to record and review their fitness information over time and compare their previous **fitness status** with their current status.

The application provides a **useful, responsive, and user-friendly interface** for displaying the user's information, statistics, and other fitness-related data.

The system will be developed using **HTML, CSS,** and **JavaScript** for the frontend, and **PHP** and **MySQL** for the backend.

User Requirements Definition:

- UR-01: The **user** shall be able to create an account by entering his name, email, password, and password confirmation.
- UR-02: The **user** shall be able to fill in the form about his body which contains the birthdate, height, weight, gender, and specifies the goal and the activity level.
- UR-03: The **user** shall be able to sign-in to the system using his email and password (no OTP).
- UR-04: The **user** shall be able to see card info (like BMI) and all sections clearly in the home page which are body, reports/statistics, exercises schedule, diet, profile, language button, and logout button.
- UR-05: The **user** shall be able to swap between the English and Arabic languages.
- UR-06: Within the **reports page**, the user shall be able to see a card of his current status which are: weight, height, BMI, SMM, PBF, and BFM.
- UR-07: Within the **reports page**, the user shall be able to see calories consumed comparing to the burned ones for the current day.
- UR-08: Within the **reports page**, the user shall be able to see his calories burned evolution for everyday within the current month.
- UR-09: Within the **reports page**, the user shall be able to see his steps which are going to be entered by the user into exercises page for every day within the current month.
- UR-10: Within the **reports page**, the user shall be able to see the changing of his weight for a duration specifies by the user.
- UR-11: Within the **reports page**, the user shall be able to export the report into a PDF file.
- UR-12: Within the **diet page**, the user shall be able to choose between recommended diet and a custom diet.
- UR-13: Within the **diet page**, the user shall be able to add, edit, and delete a meal.
- UR-14: Within the **diet page**, the user shall be able to see a bar showing his Proten, Carbs, and Fats comparing to the recommended ones.

- UR-15: Within the **diet page**, the user shall be able to see his daily calories consumed compared to the target calories consumed.
- UR-16: Within the **diet page**, if the user chooses to add or edit a meal, he must fill in the form which contains the meal name, Protein, Carbs, Fats, and Calories.
- UR-17: Within the **exercises page**, the user shall be able to choose between recommended exercises and custom exercises.
- UR-18: Within the **exercises page**, the user shall be able to add, edit, and delete an exercise.
- UR-19: Within the **exercises page**, the user shall be able to see his exercises with the ability to see and edit the weights he lifts (kg or lb).
- UR-20: Within the **exercises page**, the user shall be able to see his burned calories ~approximately, according to the count of the exercises.
- UR-21: Within the **exercises page**, if the user chooses to add or edit an exercise, he must fill in the form which contains the exercise name and choose the days of training this exercise.
- UR-22: Within the **profile page**, the user shall be able to see and edit his name, gender, age, height, weight, target weight, goal, and activity level.
- UR-23: Within the **profile page**, the user shall be able to see the date of creating his account.
- UR-24: Within the **profile page**, the user shall be able to change his password by entering the old password and then the new one.
- UR-25: Within the **profile page**, the user shall be able to delete his account.

Functional Requirements:

FR-01: Account Registration.

The system shall allow the user to create an account using his name, email, password, and password confirmation.

FR-02: Profile Creation.

The system shall allow the user to setup his body profile by entering his birthdate, height, weight, gender, goal, and activity level.

FR-03: User Authentication.

The system shall authenticate the user using his email and password without requiring an OTP.

FR-04: Dashboard Display.

The system shall display the user's body information such as BMI and weight. The navigation bar must contain the site logo, reports, exercise schedule, diet, profile, language change option, and logout option.

FR-05: Language Switching.

The system shall allow the user to switch between English and Arabic.

FR-06: Status.

The system shall display the user's status which are weight, height, BMI, SMM, PBF, and BFM.

FR-07: Daily Calorie Comparison.

The system shall display the user's calories consumed compared to calories burned for the current day.

FR-08: Monthly Calories Report.

The system shall display the user's burned calories for each day of the current month.

FR-09: Monthly Steps Report.

The system shall display the user's number of steps for each day of the current month.

FR-10: Weight Progress Report.

The system shall display the user's weight changes over a duration specified by the user.

FR-11: PDF Report Export.

The system shall allow the user to export his report as a PDF file.

FR-12: Diet Selection.

The system shall allow the user to choose between a recommended diet and a custom diet.

FR-13: Meals Management.

The system shall allow the user to add, edit, and delete meals.

FR-14: Macronutrient Comparison.

The system shall display the user's protein, carbohydrates, and fats compared to the recommended amounts.

FR-15: Daily Calorie Target.

The system shall display the user's daily calories consumed compared to his target calorie intake.

FR-16: Meal Information.

The system shall require the user to provide the meal name, protein, carbohydrates, fats, and calories when adding or editing a meal.

FR-17: Exercise Selection.

The system shall allow the user to choose between recommended exercises and custom exercises.

FR-18: Exercise Management.

The system shall display the user's exercises and allow him to view and edit the weight used for each exercise.

FR-19: Burned Calories Calculation.

The system shall calculate an approximate number of calories burned based on the count of the user's exercises (not the type of the exercise).

FR-20: Exercise Information.

The system shall require the user to provide the exercise name and training days when adding an exercise.

FR-21: Profile Management.

The system shall allow the user to view and edit his name, gender, age, height, weight, target weight, goal, and activity level.

FR-22: Account Creation Date.

The system shall display the date that the user's account was created.

FR-23: Password Change.

The system shall allow the user to change his password after verifying his old password.

FR-24: Account Deletion.

The system shall allow the user to delete his account.

Non-Functional Requirements:

NFR-01: Security.

The system shall store user passwords securely using a password hashing mechanism.

NFR-02: Data Privacy.

The system shall ensure that each user can access and modify only his own personal and fitness data.

NFR-03: Input Validation.

The system shall validate all user inputs before storing them in the database.

NFR-04: Database Integrity.

The system shall maintain the integrity and consistency of user, meal, exercise, and fitness data.

NFR-05: Performance.

The system shall respond to normal user requests within 3 seconds.

NFR-06: Usability.

The system shall provide a clear and consistent user interface that allows users to navigate between the main sections easily.

NFR-07: Language Support.

The system shall provide consistent support for both English and Arabic throughout the user interface.

NFR-08: Responsive Design.

The system shall provide a responsive interface that can be used on desktop, tablet, and mobile screen sizes.

NFR-09: Compatibility.

The system shall function correctly on commonly used modern web browsers.

NFR-10: Data Persistence.

The system shall preserve the user's account and fitness data between sessions unless the user deletes the account or the data.

NFR-11: Maintainability.

The system code shall be organized into separate files for it to be as clear as possible.

NFR-12: Error Handling.

The system shall display clear error messages when invalid data is entered or when an operation cannot be completed.

System architecture:

Frontend (HTML, CSS, JS) {

- Welcome **Page**,
- Sign-in **Form**,
- Register **Form**,
- Setup Profile **Form** (just the 1st time),
- Dashboard (Home **Page**),
- Reports **Page**,
- Diet **Page**,
- Foods adding **Form**,
- Exercises **Page**,
- Exercises adding **Form**,
- Profile **Page**

} => Backend (PHP) {

} => Database (MySQL) {

- Users,
- Profiles,
- Weight History,
- Deits,
- Foods,
- Exercises,
- Exercise History

}

Frontend:

- Pages:

1. Welcome Page:

It contains a very simple overview of the site, and a clear attractive button to sign-in.

○ Sign-in Form (separate page):

It shall ask for the email and the password from the user. It must contain a “remember me” option and a “Don’t have an account? Create now” option.

○ Register Form (separate page):

It shall ask for the name, email, password and password confirmation. Then transfer the user to the setup form.

▪ Setup Profile Form:

It shall ask for the birthdate, height, weight, gender, goal (Lose weight, Gain weight, or Maintain weight), and activity level. Then transfer the user directly to the Home Page.

2. Home Page:

It is the main dashboard which will contain the navigation-bar, the main content, the footer.

○ Navigation-bar:

It will contain the sections of the site, which are the site logo, home, reports, exercises, diet,

profile, language change button, and logout button.

- **Logout Form:**

- It must show a form like the sign-in form, which asks for the email and the password.

- **Main Content:**

- It shall display the picture of the account (it will be just a static picture with “FT” inside it) with the name, Weight, BMI, PBF, SMM, and today exercises.

- **Footer:**

- It shall show the rights of the team “Group(1)”, and when clicking on it a dropdown menu appears which contains the members names, Linked in, and GitHub.

3. Reports Page:

It must display the user’s status which are weight, height, BMI, SMM, PBF, and BFM.

It also shall show the user's calories consumed compared to calories burned for the current day, the user's burned calories for each day of the current month, the user's number of steps for each day of the current month, and the user's weight changes over a duration specified by the user. All should be showed be shown as simple graphs.

It must contain an “export to pdf” button as well.

4. Diet Page:

It shall give the user the option to choose between a recommended diet and a custom diet.

If the user chooses the recommended one, automatically he will get some meals with their amount of the protein, carbohydrates, and fats. The user can add his own meals, edit them and delete any meal he doesn't want.

If the user chooses the custom diet, he will be transferred directly to the foods adding form.

If the user clicks on a meal, he will have the choice to delete it or edit it.

There will be only one button for adding a meal which will transfer the user to the meal adding form.

It shall display the user's daily calories consumed compared to his target calorie intake

It shall display the user's protein, carbohydrates, and fats compared to the recommended amounts.

- **Foods Adding Form:**

It shall require the user to provide the meal name, protein, carbohydrates, and fats.

5. Exercises Page:

It shall give the user the option to choose between a recommended schedule and a custom schedule.

If the user chooses the recommended one, automatically he will get some exercises with the least weight. The user can add his own exercises, edit, and delete any exercise he doesn't want.

If the user chooses the custom schedule, he will be transferred directly to the exercises adding form.

If the user clicks on an exercise, he will have the choice to delete it or edit it with its weight.

It shall display each exercise with the weights that the user use, filtering by the current day or a day specified by the user.

There will be only one button for adding an exercise which will transfer the user to the exercises adding form.

It shall display an approximate number of calories burned based on the count of the user's exercises (not the type of the exercise).

- **Exercises Adding Form:**

- It shall require the user to provide exercise name and training days when adding an exercise.

6. Profile Page:

It shall show the user's account picture (it will be just a static picture with "FT" inside it, -it would be unchangeable), name, gender, age, height, weight, target weight, goal, and activity level.

It also shall display the date that the user's account was created in.

It shall show a button that gives the user the option to change his password after verifying his old password.

It shall show a button that gives the user the option to delete his account.

- **Delete Account Form:**

It shall require the user to enter his email and the password with the password confirmation, with the buttons of “Delete” and “Cancel.”

Clicking the button “Delete” will give a popup to ask the user if he is sure about deleting the account.

- **Notes:**

- You don't have to work exactly as written above; it is just a guide if you don't know what to do.
- Everyone has to work only on his pages folders.
- I assume everyone already knows how to work with git and GitHub, so in GitHub there will be branches, each with a person name. Your work must be there and then if you done from your work do a pull request. All of this is to not cause a conflict.
- The overall theme will be unified in the global.css to not be ugly.

- To test your work, I suggest using the extension “live server” in vs code.
- The README.md will contains the project structure.

Tasks to do:

NO	Task	Name	Dou date	Status
T1	SRS	Reda	22 th Sep	
T2	Project structure	Reda		
Frontend				
T3	Frontend – home page	Reda	25 th Oct	
T4	Frontend – welcome page	Reda		
T5	Frontend – sign-in form	Mohammed	25 th Oct	
T6	Frontend – register form	Mohammed		
T7	Frontend – profile page	Mohammed		
T8	Frontend – reports page	Hassan	25 th Oct	
T9	Frontend – diet page	Redwan	25 th Oct	
T10	Frontend – exercises page	Ahmed	25 th Oct	
Backend				
T11				
T12				
T13				
T14				
T15				
T16				
T17				
T18				